

LKE Standard

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Prepared by: Bryan Woo

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Revision History

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1.0 Introduction of LKE Standard

The LKE Standard represents an integrated approach to modern durian cultivation, combining traditional agricultural expertise with cutting-edge technology to ensure sustainable and efficient practices. It is built upon two key frameworks: the **LKE Durian Planting Framework** and the **Smart Farming Framework**. Together, these frameworks form the backbone of our cultivation strategy, enabling comprehensive management and optimization across all aspects of durian farming.

Within the overarching framework, utilizing IoT sensors, advanced analytics, and automated systems. Together, the LKE Durian Planting Framework and the Smart Farming Framework provide a robust, adaptable standard that drives innovation and sustainability in the durian industry.

LKE Standard serves as a guideline to streamline the entire durian cultivation process. By adopting this standard, durian planters can benefit from structured workflows and advanced tools, which provide valuable insights and practices to enhance efficiency and sustainability in durian farming. While the framework serves as a reference, its application should be adapted to individual orchard conditions for optimal results

In the following sections, we will explore these frameworks in greater detail, highlighting their sub-components and how they contribute to the overall efficiency and sustainability of durian cultivation.

2.0

LKE Durian Planting Framework

2.1 Suitability of Planting Location

The suitability of a location is a critical factor in ensuring the success of durian cultivation. This document provides scientifically recommended parameters to guide the selection of planting sites, emphasizing the importance of environmental and soil conditions. These parameters represent optimal ranges for growth and productivity, serving as a reference for decision-making.

2.1.1 Recommended Benchmark Parameter

Parameter	Recommended Range	Importance
Rainfall	2000-3000 mm	Provides essential moisture for tree hydration and supports healthy flowering and fruiting.
Temperature	27°C - 32°C	Ensures consistent growth; extremes can stress trees and reduce yields.
Humidity	80% - 90%	Critical for flowering and fruiting; low humidity may cause poor fruit development.
Slope	0° - 20°	Ensures proper drainage and minimizes waterlogging risks.
Sea Level	50 - 400 m	Suitable for tropical conditions required by durian trees.
Soil Type	Well structured & textured clay to loam	Supports root development and retains essential nutrients.
Soil pH	5.5-6.5	Optimal for nutrient availability; acidic or alkaline soils may require amendments.
Soil Salinity	< 1 dS/m	Low salinity ensures nutrient uptake and avoids soil toxicity.

Practical Considerations: While this section 2.1 provides specific ranges, real-world scenarios often require balancing trade-offs between different factors.

While this document provides scientifically recommended ranges, some factors hold greater importance due to their impact and level of controllability.

2.1.2 Key Considerations

A. Rainfall (A Non-Negotiable Factor)

- Rainfall is one of the most critical parameters because it cannot be artificially controlled or adjusted. Locations with rainfall within the recommended range of 2000–3000 mm annually are highly preferred, as they ensure sufficient water for tree hydration, flowering, and fruiting.

B. Soil and Terrain (Factors That Can Be Improved)

- Parameters such as soil type, pH, salinity, and drainage can often be corrected through soil treatments or engineered solutions:
- Soil pH: Adjusted with lime (to increase pH) or sulfur (to lower pH).
- Soil Salinity: Reduced through soil washing or the application of gypsum.
- Drainage: Improved with engineered drainage systems or reshaping the terrain.
- Soil Fertility: Enhanced through the addition of organic matter, compost, or fertilizers.

C. Sea Level and Slope (Minimal Concern)

- Sea level and slope are less critical factors compared to rainfall and soil parameters. Flat terrains or low-altitude locations can be optimized through proper design and engineering, such as implementing terracing or ensuring adequate water flow.

2.1.3 Case Study Example

When comparing two potential locations:

Site A: Receives rainfall within the ideal range but has poor soil quality.

Site B: Has better soil quality but excessive rainfall beyond the recommended range.

In this scenario, **Site B** would be the preferred choice because soil quality is the most critical factor for long-term productivity. Even though Site B receives excessive rainfall, this issue can be managed by implementing an effective drainage system. In contrast, poor soil quality at Site A would pose a more significant challenge, as degraded soil limits plant growth and may require extensive remediation efforts that are not always successful.

This highlights the importance of prioritizing fundamental factors like soil quality over environmental conditions that can be controlled through infrastructure improvements. By selecting a site with better soil structure and fertility, long-term sustainability and productivity can be ensured, even in less-than-ideal climatic conditions.

2.1.4 Water Source

A. Mountain water

Originates from highland springs and natural catchments, often filtered through rocks and soil, resulting in high purity and oxygen-rich water.

Advantages:

- High dissolved oxygen (DO) levels, promoting better root respiration and nutrient absorption
- Naturally filtered, low in sediments and pollutants
- Rich in minerals, which may benefit soil health
- Consistent flow in well-preserved ecosystems
- Lower risk of industrial or agricultural contamination

Considerations:

- Seasonal variations in water availability (may decrease in dry seasons)
- Requires proper storage and distribution to ensure continuous supply

B. River water

Flows through various landscapes and is influenced by rainfall, upstream activities, and environmental factors. Its dissolved oxygen levels vary, depending on water flow and contamination.

Advantages:

- Higher availability, supporting large-scale irrigation
- Easier to access in certain regions
- Can support high irrigation demands in intensive farming

Disadvantages:

- Lower dissolved oxygen (DO) levels in stagnant or slow-moving sections, which may affect plant root respiration
- Higher sediment levels, requiring filtration before use
- Risk of contamination from upstream agricultural, industrial, or urban activities
- Water quality can fluctuate seasonally due to rainfall and pollutants

2.2 Pre-Planting Operation

2.2.1 Land Clearing

Before planting durian trees, thorough land clearing is essential to ensure a healthy and disease-free growing environment. This process involves the complete removal of previous vegetation, including oil palm, rubber trees, old orchards, or secondary forests, using excavators and other heavy machinery. A well-executed land clearing process eliminates potential threats to durian trees and provides an optimal foundation for long-term orchard productivity. It ensures: A clean and disease-free soil environment for young durian trees. Better pest management by removing old tree residues that attract harmful insects. Improved soil aeration and structure, promoting healthy root growth. Efficient drainage, reducing the risk of root rot and other water-related diseases.

A. Complete Removal of Previous Crops

If the land was previously cultivated with oil palm or rubber trees, the clearing process must be extremely thorough. All roots, stumps, and residual plant materials must be completely removed to prevent the risk of disease carryover and pest infestation.

- **Disease Prevention:** Old tree residues may harbor fungal pathogens, bacterial infections, and harmful nematodes that can affect young durian trees.
- **Pest Control:** Leftover roots and decomposing plant matter can attract ants, termites, and other harmful insects, which may damage durian saplings.

B. Soil Preparation and Deep Tillage

After vegetation removal, the soil undergoes deep tillage and leveling to enhance soil structure and fertility. This process includes:

- Turning over the soil to expose and remove hidden pests and pathogens.
- Breaking up compacted soil layers to improve root penetration and drainage.
- Ensuring proper land grading to facilitate efficient water flow and prevent waterlogging.

2.2.2 Infrastructure

A. Plantation Mapping & Planning

Plantation mapping is a critical step in establishing and managing an orchard. By utilizing GPS marking tools, we can accurately define the boundary of the plantation, ensuring clear land ownership and efficient planning for infrastructure development.

- **Precise Boundary Identification:** Using GPS tools to map the plantation border helps clearly define the area, avoiding land disputes and ensuring proper land utilization.
- **Accurate Distance Measurement:** Mapping allows us to measure distances within the plantation, aiding in the planning of internal infrastructure such as farm roads, irrigation systems, and planting rows.
- **Facilitates Fencing and Gate Installation:** With precise border data, installing fencing and positioning entry gates becomes much more efficient and accurate, improving security and accessibility.
- **Supports Future Expansion and Planning:** A well-documented plantation map serves as a reference for future expansion, zoning, and land development, ensuring systematic growth and efficient resource management.



Infrastructure includes essential components such as farm roads, ponds, terracing, fencing, and accommodation offices.

Beyond these basics, we have integrated Smart Farming technology to enhance monitoring and management. This includes weather stations, soil sensors, and LoRaWAN gateways, which provide real-time data to support precise and efficient farming practices.

These advanced tools complement our traditional infrastructure, creating a robust foundation for sustainable durian cultivation.

B. Smart Farming Hardware

i) Integrated Soil Sensor



This sensor monitors essential soil parameters, including NPK%, pH, EC, moisture, temperature and soil tension. This sensor provides real time insights into soil health, enabling data-driven adjustments to optimize growing conditions.

iii) Weather Station



This weather station captures data on temperature, rainfall, and humidity, providing essential information to support irrigation planning.

iv) LoRaWAN Gateway



This station is equipped to measure pH, EC, and turbidity, this station monitors water quality, ensuring that irrigation meets optimal standards for plant health.

v) Solar Light-Based Insect Trap



This device uses solar-powered light spectrum to attract and trap insects, reducing pest pressure and the need for chemical pesticides. It contributes to a more sustainable and environmentally friendly approach to pest control.

vi) CCTV Camera



Installed for monitoring purposes, the CCTV camera provides visual surveillance of the orchard, ensuring security and facilitating remote observation of orchard conditions.

C. Infrastructure Layout

i) Water Storage



Water storage system designed to meet irrigation demands across the orchard. This system comprises strategically placed ponds and reservoirs that provide a consistent water supply during dry periods.

ii) Farm Roads



Farm roads structures to facilitate efficient movement and access for the transportation of equipment, supplies, and harvested fruit, enhancing operational efficiency and supporting maintenance activities.

iii) Sprinkler Irrigation



Each sprinkler emits the exact same amount of water. This practice helps maintain uniformity and consistency across the entire farm, ensuring that every tree receives the same level of moisture and nutrients.

iv) Fencing and Gate



The completion of fencing, gate, and overall enclosure installation signifies a proactive step in safeguarding our company's assets. Meticulously installed, these measures, along with the installation of CCTV cameras across the farm, serve to protect against potential threats, including wildlife intrusion or unauthorized access, thereby ensuring the integrity and security of our orchard.

v) Drainage



This drainage system is designed to efficiently manage water flow and prevent water accumulation. The use of a covered culvert helps direct excess water away, reducing the risk of erosion and ensuring stable ground conditions. This system enhances overall water management in the orchard, especially during heavy rainfall.

vi) Store Room



The store-room must be clean and organized. Proper labelling of agrochemicals ensures safety and support Good Agricultural Practices (GAP).

vii) Pump Room



A clean and well-maintained pump room ensures smooth irrigation operations and prevents system malfunctions.

D. Specification of Main Road

The main road consists of a 5.0-meter wide carriageway with 0.6-meter compacted shoulders on both sides. It is flanked by drainage ditches measuring 0.6 meters in width and 1.8 meters in depth. Trees are planted beyond the drainage zone for landscape and erosion control purposes.



E. Specification of Sub Road

The sub road features a 4.0-meter wide carriageway with 0.3-meter compacted shoulders on both sides. Drainage ditches are 0.3 meters wide and located before a 1.4-meter tree buffer zone, providing basic accessibility and field connectivity.



F. Specification of Terracing



- Terraces are constructed with a width ranging from 4.2 to 5.5 meters.
- Each terrace is sloped outward with internal drainage to manage runoff effectively.
- Terrace construction is done after road development.
- Terrace is designed to allow access from front and exit from the back, ensuring smooth movement across the terrace without needing to turn back.
- Conservation terraces are required depending on the slope gradient to reduce water runoff speed and prevent erosion.

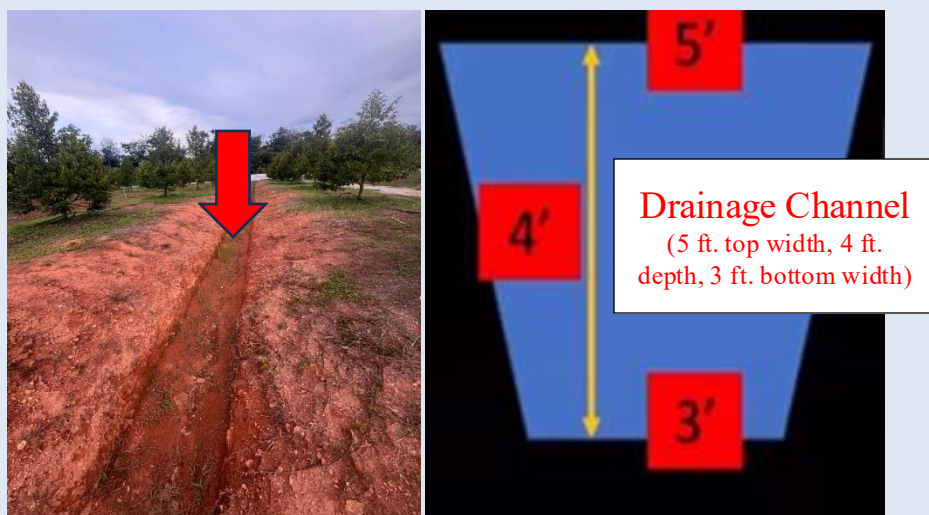
For slopes less than 5° (1:12), at least 1 conservation terrace is needed for every 12 normal terraces.
For gentle slopes (5°–8°, 1:12–1:7), 1 conservation terrace is recommended for every 7 to 12 terraces.
For steep slopes (8°–14°, 1:7–1:4), 1 conservation terrace is required for every 4 to 7 terraces.

Remark: Specification of terracing depends on the slope and topography

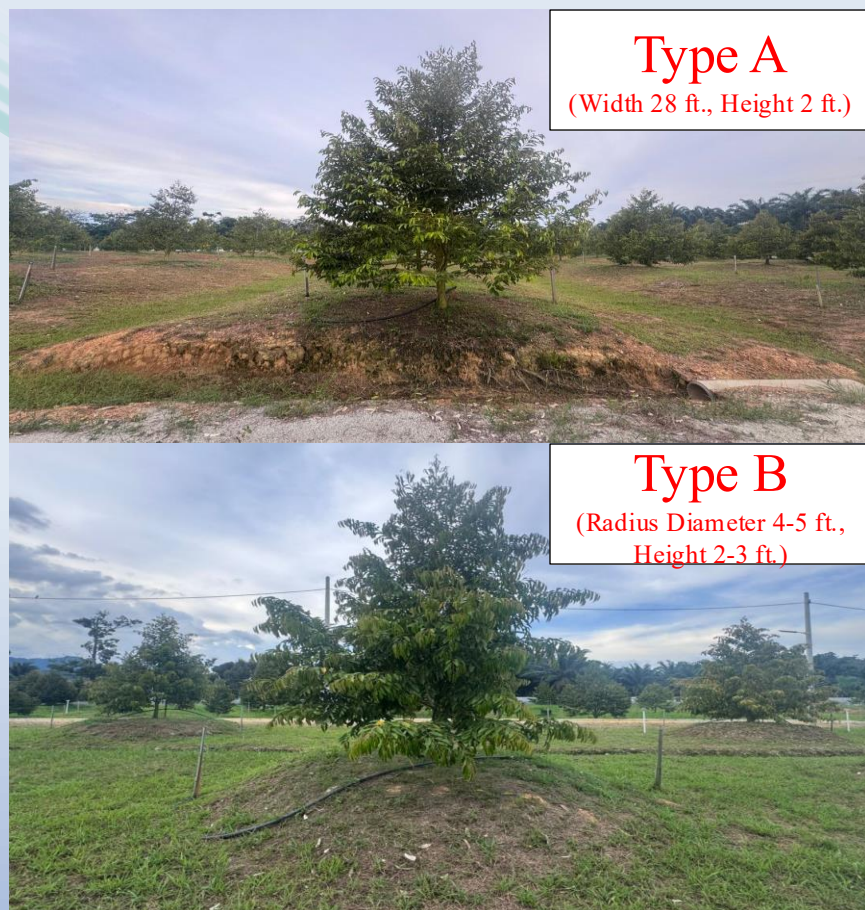
2.2.3 Raised Beds & Drainage System

To minimize waterlogging risks, improve soil drainage and enhance root zone aeration for durian trees by integrating drainage channels with raised bed structures.

The following diagram shows drainage channels in the LKE durian orchard.



The following diagram shows two types of raise beds in the LKE durian orchard.



2.2.4 Irrigation

To ensure a standardized irrigation method is established in the LKE durian orchard, sprinkler irrigation is implemented to provide consistent and efficient water distribution. This sub-stage focuses on outlining the optimal placement and installation of sprinklers around durian trees to maximize water efficiency and support healthy tree growth. By following these guidelines, each tree receives adequate water to meet its daily requirements, promoting sustainable orchard management.

A. Pumps for Irrigation & Trouble-Free Operation of Centrifugal Pump

We use centrifugal pump to efficiently move water throughout the orchard. This pump uses rotating impellers to increase water pressure, allowing smooth flow through the irrigation system, powered by diesel engines. To ensure trouble-free operation of centrifugal pump in our irrigation system, we recommend two key requirements as follows.

- Cavitation can lead to severe damage within pump, affecting its efficiency and lifespan. It is essential to prevent cavitation under all operating conditions.
- A steady minimum flow must be maintained to protect the pump from issues such as overheating and excessive wear. This helps extend the pump's service life and ensures consistent water delivery.

B. Key Performance Parameter of Centrifugal Pump

To ensure efficient and reliable operation of centrifugal pumps in our irrigation system, the following key performance parameters must be monitored and maintained.

Capacity:

- Capacity means the flow rate with which liquid is moved or pushed by the pump to the desired point in the process.
- It is commonly measured in either gallons per minute (gpm) or cubic meters per hour (m³/hr).

Head:

- The term “head” is used instead of “pressure” to describe the pump's performance.
- Head represents the height to which the pump can raise a liquid, based on the energy imparted by the pump to the liquid.
- This is measured in feet or meters of liquid column and is not influenced by the liquid's specific gravity (weight).

C. Sprinkler Nozzles

We select sprinklers based on three key factors as follows.

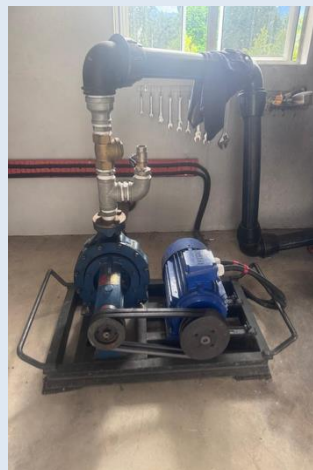
- **Distance of Throw:** This refers to the radius of water distribution from the sprinkler. We choose sprinklers with a throw distance that allows for optimal coverage around each durian tree, ensuring water reaches the root zone effectively.
- **Application Rate:** Also known as the precipitation rate, this measures how quickly water is applied to the soil. We select sprinklers with an application rate suited to the soil's absorption capacity, minimizing runoff and ensuring efficient water usage.
- **Distribution Uniformity (DU):** This factor indicates how evenly water is applied over the area. High distribution uniformity is essential to prevent overwatering in certain zones and underwatering in others, promoting consistent soil moisture levels across the orchard.

D. Sprinkler Placement

To ensure even water distribution and efficient coverage, the placement of sprinklers around durian trees follows specific guidelines.

- **Distance from Trees:** Each sprinkler should be placed at a distance that maximizes root zone coverage without causing direct spray on the tree trunk. This distance may vary depending on tree age and growth stage.
- **Spacing Between Sprinklers:** Sprinklers are spaced to create overlapping coverage zones, reducing dry areas and promoting consistent soil moisture levels across the orchard.
- **Installation Height:** Sprinklers should be installed at an appropriate height to ensure optimal spray trajectory and prevent water blockage by surrounding vegetation.
- **Orientation and Angle:** Each sprinkler is positioned at an angle that minimizes wind drift and directs water toward the base of the trees.

The following diagram provides an overview of the pump specifications used in LKE durian orchard.



BRAND: UNOFLOW
MODEL: XA32/26
IMP. Size: 264 MM
SPEED: 2900 RPM
CAPACITY: 22 M/HR
HEAD: 95 M
POWER: 18.5 KW
SERIAL NO:
20201200

The following diagram shows two types of sprinkler nozzles used in the LKE durian orchard.



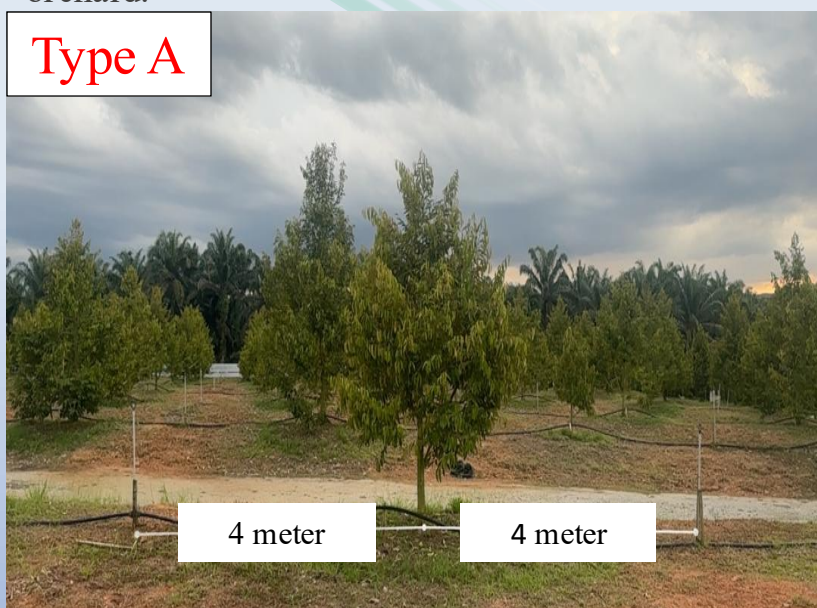
PC Sprinkler Thread ¾ "



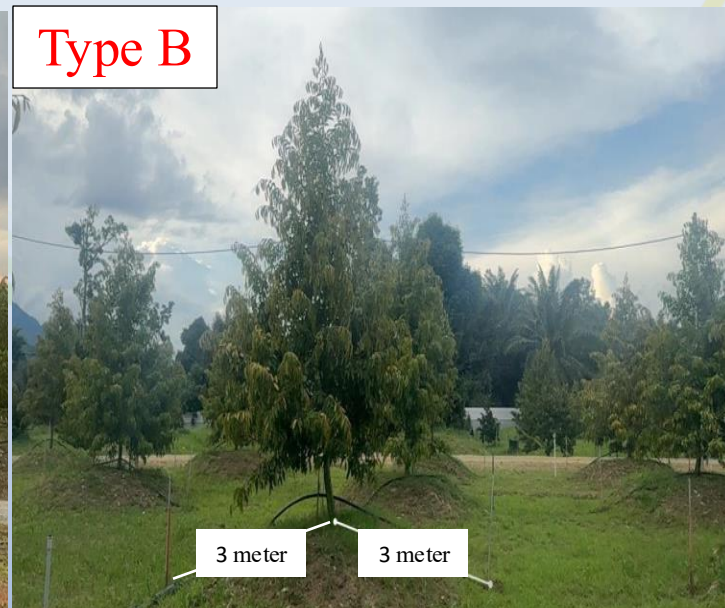
Female Thread ¾ "
Butterfly Sprinkler

The following diagram illustrates two types of sprinkler placements implemented in LKE durian orchard.

Type A



Type B



2.2.5 Water Reservation

To ensure that sufficient water reserves are maintained to support orchard needs during extended dry periods. The total amount of water required for storage is based on calculations from **section 2.4.1 Watering**, which outlines the water needs per tree per day. This sub-stage provides formula for calculating the necessary pond size or water storage capacity, ensuring that water reserves meet orchard requirements during periods without rainfall.

A. Total Water Required to Reserved

The total water required to reserve per acre (D) is calculated using the water required per tree per day (A) for different tree age (Year 1-10), water required per tree per month (B) and tree planted per acre (C).

The calculation is as follows:

$$D = A \times B \times C$$

where:

- A = Water Required per tree per day (as specified in the **section 2.4.1.1 Watering** based on tree age)
- B = Watering Frequency per month (3 days Watering Interval)
- C = Number of trees per acre

The following diagram provides an overview of total water required to reserved, based on tree age and other estimated factors:

Tree Age	(A) Water Required per tree per day (L)	(B) Watering Frequency per month (3 days Watring Interval)	(C) Tree Planted per acre	(D) Total Water Required to Reserved per acre (L)
1	30	10	40	12,000
2	60	10	40	24,000
3	90	10	40	36,000
4	120	10	40	48,000
5	150	10	40	60,000
6	180	10	40	72,000
7	210	10	40	84,000
8	240	10	40	96,000
9	270	10	40	108,000
10	300	10	40	120,000

B. Design of LKE Farm Pond

Layered Trapezoidal Pond, this design uses a 4 layered structure, but the formula can be extended to any number of layers to meet different water storage capacity requirements.

The calculation is as follows:

$$V = (A^2 + (A+B)^2 + (A+2B)^2 + (A+3B)^2) \times C$$

Where:

- V = Total volume (m^3)
- A = Side length of base layer
- B = Increment in side length for each successive layer
- C = Height of each layer

i) Case Study Example: LKE Farm Pond Design

In this case study, the LKE Farm Pond was designed to meet a water storage requirement for one month correspond to 14 acres land with a total of 560 durian trees.

Pond Specifications

- **Base Layer (A):** 28 m $\times$ 28 m
- **Increment per Layer (B):** 5 m
- **Number of Layers:** 4 layers
- **Layer Height (C):** 1 m

Volume Calculation: Using the layered trapezoidal pond formula:

$$V = (A^2 + (A+B)^2 + (A+2B)^2 + (A+3B)^2) \times C$$

Substitute the values:

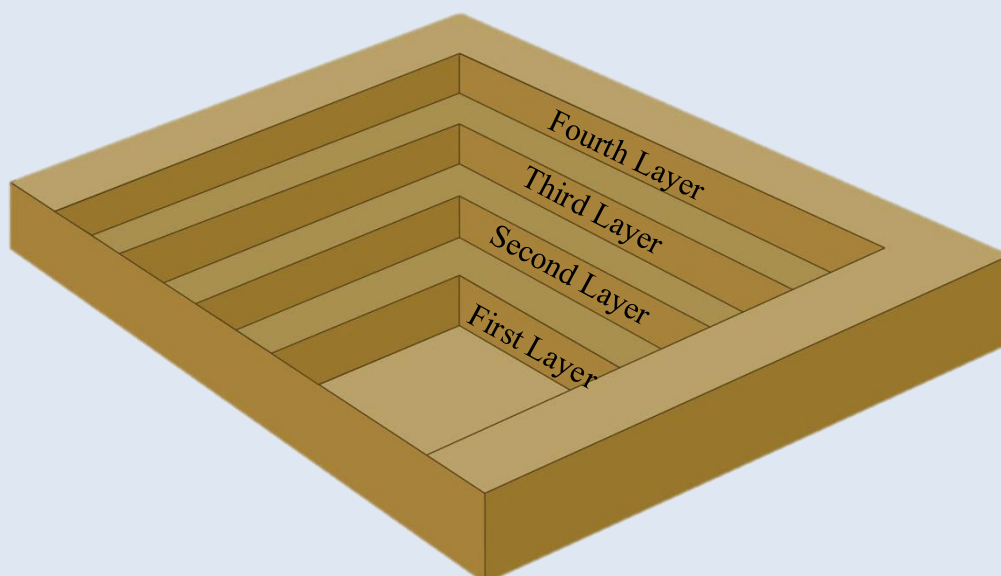
$$V = (28^2 + 33^2 + 38^2 + 43^2) \times 1$$

$$V = (784 + 1089 + 1444 + 1849) \times 1$$

$$V = 5166 m^3$$

$$V = 5 \text{ million liters of water}$$

To complement the calculation formula, the following images provide a visual representation of the LKE Farm Pond's design.



These visuals demonstrate how the theoretical design principles are effectively implemented in real world scenarios. This section provides a holistic understanding of how the pond supports water reservation and irrigation needs, contributing to sustainable durian cultivation.



2.3 Seedling Planting

2.3.1 Source & Criteria of Seedlings

This section provides guidelines for selecting high-quality durian seedlings that meet specific criteria to ensure successful orchard establishment. It emphasizes sourcing seedlings from certified nurseries, verified by the Department of Agriculture, to guarantee compliance with quality standards.

A. Selection Criteria of Durian Seedlings

- Select seedlings with healthy lateral buds that grow actively.
- Choose seedlings with a balanced leaf count; avoid those with overly large or small leaves. (20-30 leaves glowing leaves without black spots)
- Ensure seedlings have a height of 2-3 feet and the main stem must present brown colour.
- Verify seedlings are free from disease and nutrient deficiencies.
- Ensure the main stem is sturdy and upright.
- Select seedlings with a thick base to promote nutrient storage and new shoots.
- Avoid seedlings with roots protruding from the polybag to prevent stunted growth.

The following diagram show examples of durian seedlings that meet the recommended selection criteria, including healthy lateral buds, balanced leaf count, and appropriate stem diameter and height.



B. Source of Durian Seedlings

- Ensure to purchase recommended durian saplings from agencies or nurseries that have obtained the official verification stamp from the Department of Agriculture.
- This stamp confirms that the nursery meets the required standards for seedling quality.

The following diagram shows example of a sample image of Department of Agriculture's verification stamp


 ABT 8

PERAKUAN PENGESAHAN BAHAN TANAMAN

1. Nama Peserta Skim : LONG YOU NURSERY SDN BHD

2. No. Pendaftaran Peserta Skim : N(J) 6

3. Alamat Ladang/Nurseri : BATU 1 ½, JALAN BAKRI, 84000 MUAR, JOHOR

4. Tarikh Pemeriksaan : 16/4/2024

Dengan ini diperakukan bahawa bahan tanaman ini adalah tulen dan berkualiti serta memenuhi Standard Minimum Bahan Tanaman yang ditetapkan di bawah Skim Pengesahan Bahan Tanaman, Jabatan Pertanian Malaysia.

Bil.	Jenis Tanaman	Nama Varieti/No. Kod Pendaftaran	Tahun Tanam	No. Blok Pokok Induk	Bilangan Bahan Tanaman Diperakukan	No. Pendaftaran Bahan Tanaman
2	Durian	D197 (mk)			43 623	N(J)6/2024/01/00001 - 43623
		D200 (Cochi)			6990	N(J)6/2024/01/0001 - 6990
		D168 (Hjh. Hasmah)			15713	N(J)6/2024/01/0001 - 15713
		D175 (udang Mark)			126	N(J)6/2024/01/0001 - 126

Tandatangan:


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Tandatangan:


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 86100 Ayer Hitam, Johor.

Cop SPBT:

Tarikh: 16/4/2024

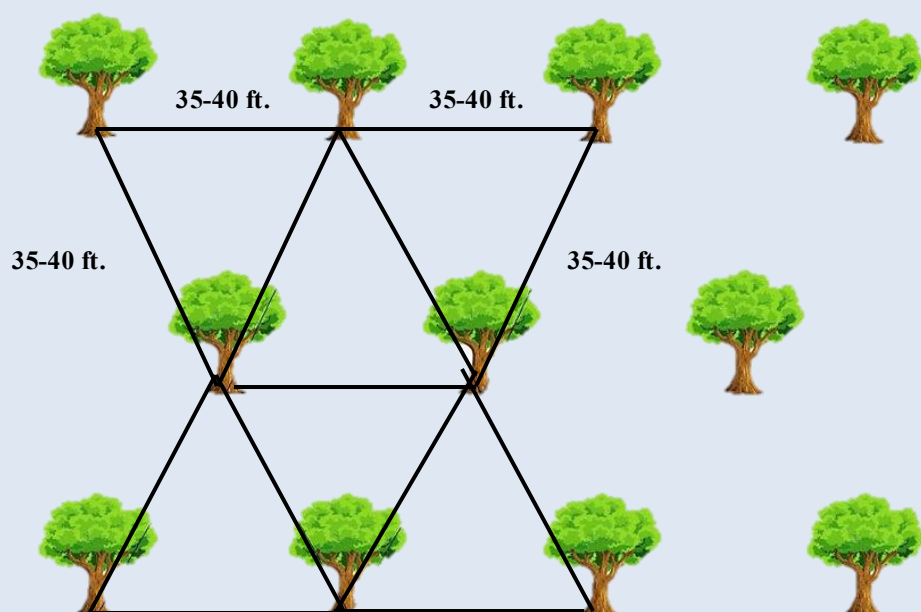


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2.3.2 Planting Density

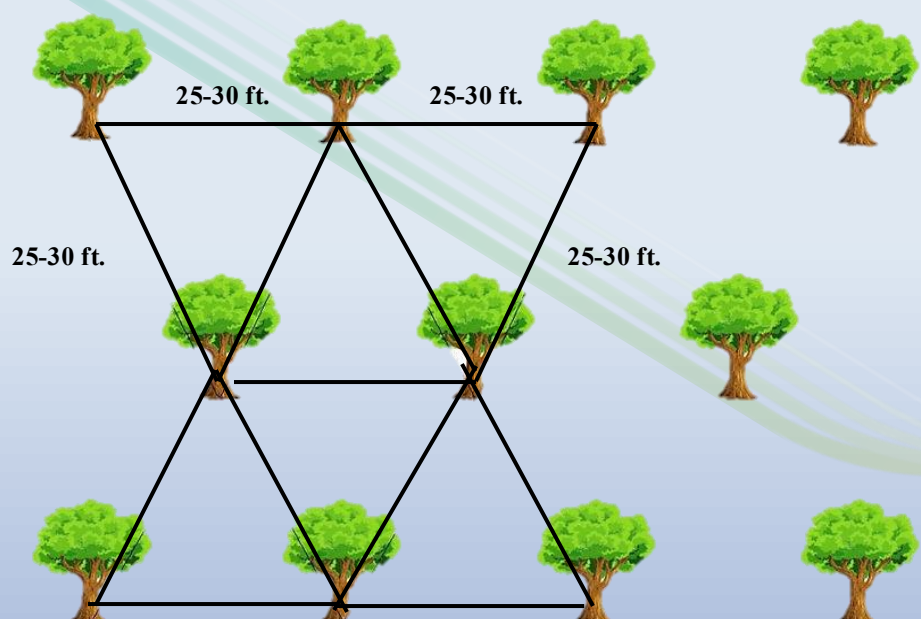
This section provides guidelines for planting density to ensure optimal land use and tree growth based on durian species such as Musang King (MK) and Black Thorn (BT).

A. Musang King Planting Distance



**Recommended MK
Stand per Acre
(SPA)
40 tree / acre**

B. Black Thorn Planting Distance



**Recommended BT
Stand per Acre
(SPA)
50 tree / acre**

2.3.3 Initial Planting of Durian Seedlings

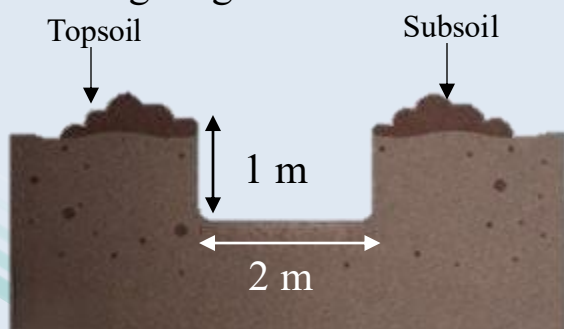
This section provides standardized guidelines for the initial planting of durian seedlings, including specifications for planting holes and the application of base fertilizer. This section applies specifically for the initial planting stage. For ongoing fertilization beyond the initial planting, refer to the [section 2.4.2 Fertilizer](#), which covers fertilization practices from year one to year ten.

A. Planting Process

i) Hole Size

- Planting holes should be excavated three to four weeks before planting.
- Dig a planting hole of specified dimensions to provide adequate space for root development.
- Holes should be 2m x 2m and 1m deep.

The following diagram shows the hole size that are in LKE durian orchard.



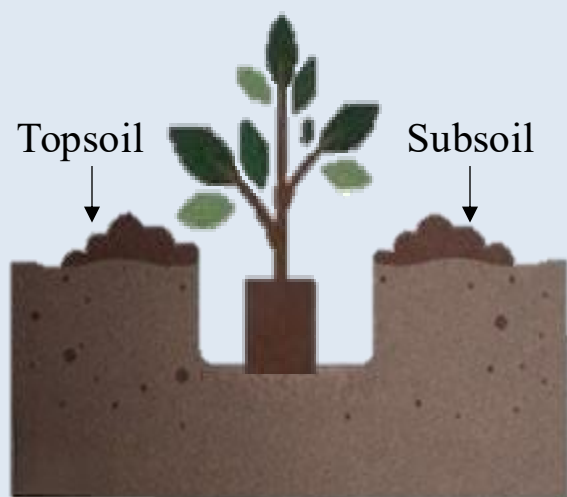
ii) Planting is best done during the onset of the rainy season

- After digging the planting hole, allow the soil to undergo natural compaction by rainfall.
- The soil should be left to settle and stabilize after at least two rainfalls.
- This process ensures that the soil is more compacted and stable, providing a solid base for planting.
- Once the soil has adequately settled after two rainfalls, proceed to the next step.

iii) Durian Seedling Placement

- Remove Polybag: Carefully remove the polybag from the seedling to avoid damaging the roots.
- Place Seedling in the Planting Hole: Gently place the seedling into the center of planting hole.

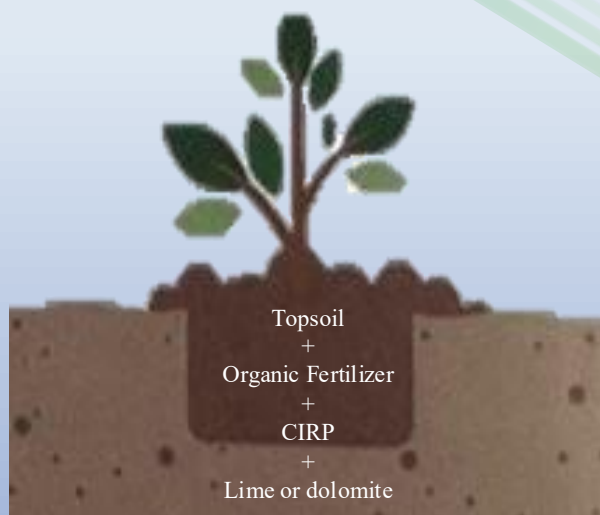
The following diagram shows the placement of durian seedling that are in LKE durian orchard.



iv) Backfilling with Soil Mixture

- Mix topsoil with organic fertilizer (recommended ratio, e.g., 3:3:3), 200g of CIRP, dolomite if the pH is lower than 5.5 and subsoil to create a nutrient-rich backfill.
- Use the prepared soil mixture that mentioned in section 3.4.1 to fill the gaps around the durian seedling.
- Ensure the soil is lightly compacted to stabilize the seedling without compressing the roots of seedling.

The following diagram shows the backfilling process that are in LKE durian orchard.



B. Post-Planting Care

Follow-Up Steps

- After initial planting, refer to the **section 2.3.4 Shade Requirement** to ensure the seedling is adequately protected from direct sunlight to support ongoing growth process.
- After initial planting, refer to the **section 2.4.1 Watering** ensure the seedling is receiving adequate amount of water to support ongoing growth process.
- After initial planting, refer to the **section 2.4.2 Fertilizer (Growing Stage)** to ensure the seedling is receiving adequate amount of nutrients to support ongoing growth process.

The following diagram shows the actual seedlings planted in the LKE durian orchard.

To enhance growth conditions and minimize weed competition, an optional weed mat can be used around the base of the seedlings. This mat helps suppress weed growth, conserve soil moisture, and reduce the need for frequent weeding, promoting healthier early development for the durian trees.



2.3.4 Shade Requirement

This section provides guidelines for the proper handling and acclimatization of durian seedlings sourced from nurseries when planting them in orchard and prevent direct sun exposure and other environmental stresses, ensuring the seedling adapt effectively to field conditions and promote healthy growth.

Application of Netting

Netting is used to protect young durian seedlings from direct sunlight and any other environmental stresses when they are first transplanted into the field. This section outlines the proper installation and removal process for netting, ensuring optimal growth conditions as the seedlings develop.

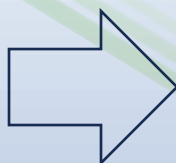
A. Net Installation

- When the durian seedlings are newly planted, a black net is installed to shield them from excessive sunlight.
- This step is crucial to help the young tree acclimatize to field conditions.

B. Net Removal

- As the tree grows and surpasses the height of the netting, it indicates that the seedling is strong enough to tolerate full sun exposure.
- At this stage, the net can be removed to allow tree unrestricted access to sunlight.

The following two diagrams show the transition of durian seedlings in the LKE durian orchard, from being covered with netting for initial protection to growing without netting as they acclimate to full sunlight.



2.3.5 Numbering & Tagging

At LKE, each durian tree is assigned a unique number for identification, marked on a metal stake positioned next to the tree and linked to an NFC tag.

This system allows workers to log activities—such as weeding, pruning, or fertilizing and etc.—by tapping on the NFC tag for each tree.

Because each tree has its own designated number, we can ensure that no tree is missed during routine tasks. If a worker skips a tree, it will be clear from the records, allowing us to verify that all tasks are completed thoroughly and accurately.

The following left diagram shows the numbering system used in the LKE durian orchard. Each tree is assigned a unique number displayed on a metal stake positioned next to it. The numbering format includes a two-letter prefix that indicates the specific orchard block or zone, ensuring precise identification and management across different areas.

The following right diagram shows the NFC tag attached to the trunk of durian tree. The functionalities of the NFC tag will be discussed in greater detail in the **section 3.0 LKE Smart Farming Framework**, where its applications in smart agricultural management will be elaborated.



2.4

Growing Stage (Before Harvesting)

2.4.1 Watering (Growing Stage)

This section provides standardized methods for calculating the water requirements of durian trees.

These calculations consider multiple factors to ensure each tree receives adequate water for optimal growth and health.

This approach is based on academically proven methods for calculating water requirements, specifically incorporating the soil factors as key variable.

A. Definition of Calculation Factors

- 1) Soil Texture: Different Soil Texture, has different value of AWC
- 2) AWC: Available Water Capacity of the soil (mm/m)
- 3) RZD: Root Zone Depth (cm), assume roots of any durian tree has the same depth according from year 1 to 10
- 4) TAW: Total Available Water (mm), derived by multiplying RZD by AWC, representing the total water available in the root zone.
- 5) ASMD: Allowable Soil Moisture Deficit (%), indicates the level of water depletion allowed before the next irrigation, assume ASMD = 50% (ASMD may vary according to decision maker).
- 6) NID: Net Irrigation Depth (mm), derived by multiplying TAW by ASMD, representing the necessary water depth per irrigation cycle.
- 7) π : The number π is a mathematical constant that is the ratio of a circle's circumference to its diameter, approximately equal to 3.14.
- 8) R: Canopy diameter (m), assume equal to tree age (e.g. Year 1 = 1 m of canopy diameter; Year 10 = 10 m of canopy diameter)

*Primary Formula of Water Required per Tree = $NID \times \text{Canopy Area}$
= $(TAW \times ASMD) \times \text{Canopy Area}$
= $(AWC \times RZD \times ASMD) \times \text{Canopy Area}$

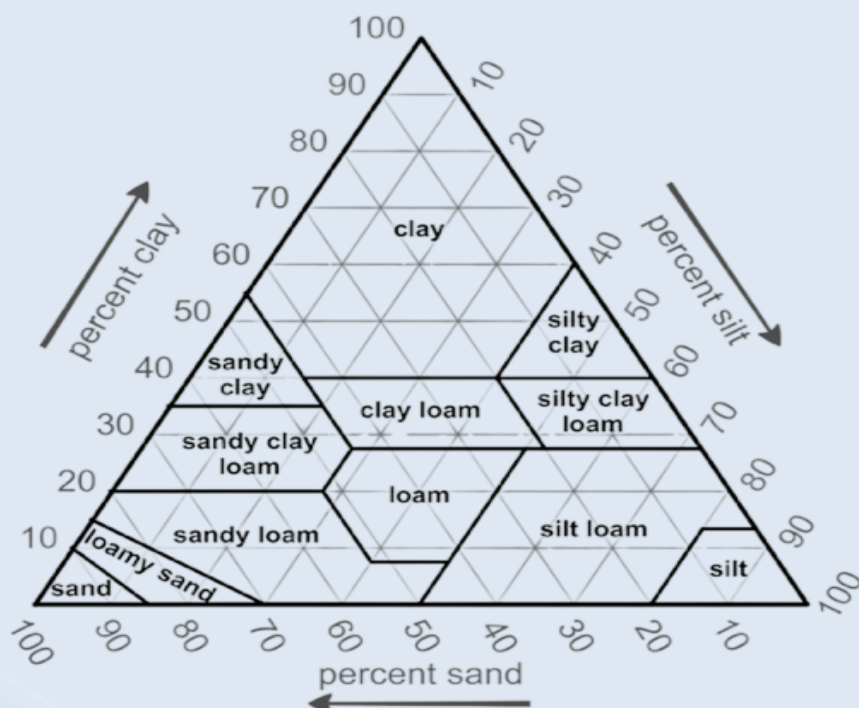
*Derived Formula 1
 $NID = TAW \times ASMD$

*Derived Formula 2
 $TAW = AWC \times RZD$

B. Step-by-step calculation with case study example

Step 1:

Determine your soil texture class by conducting soil analysis.



The left diagram shows soil texture class determined by the percentage of sand, silt and clay.

Perform soil analysis and consult subject matter expert to identify the soil texture class based on the proportion of sand, silt and clay.

Case Study:

In LKE orchard, soil analysis revealed the texture is **Sandy Loam**

Step 2:

Determine your AWC value according to your soil texture.

Texture Class	AWC (mm/m)
Coarse Sand	20 - 60
Fine Sand	60 - 80
Loamy Sand	90 - 100
Sandy Loam	104 - 117
Fine Sandy Loam	125 - 167
Silt Loam	167 - 200
Silt Clay Loam	150 - 167
Silt Clay	125 - 142
Clay	100 - 125

The left diagram shows the AWC value corresponding to their respective texture class.

In other words: Different Soil Texture, has different value of AWC

If the soil testing method is available for direct AWC measurement, this may provide even more precise values.

Case Study:

For **Sandy Loam**, the range of AWC is between 104 - 117 mm/m based on the chart. In this case, we use 104 mm/m in the following steps.

Step 3:

Determine your Root Zone Depth (RZD).

Tree Age	Estimated RZD (cm)
1	30
2	32.22
3	34.44
4	36.66
5	38.88
6	41.1
7	43.32
8	45.54
9	47.76
10	49.98

The left diagram shows the estimated root zone depth (cm) for durian tree from year 1 to 10.

Case Study:

- The soil texture has already been determined as **Sandy Loam** from Step 1, and its AWC value is 104 mm/m, as identified in Step 2.
- The orchard has durian trees at Year 3, so the corresponding RZD value from the chart is 34.44 cm.

Step 4:

Calculate Total Available Water (TAW) by multiplying RZD and AWC.

The below diagram shows the TAW value calculated by RZD and AWC from previous steps.

Soil Texture	AWC (mm/m)	RZD (m)	TAW (mm)
Sandy Loam	104	0.3444	35.82

Formula:

- $TAW = AWC \times RZD$
- RZD (cm) to RZD (m), divided by 100

Case Study:

- So we know that the TAW for the orchard is 35.82 mm.

Step 5:

Set ASMD percentage and calculate Net Irrigation Depth (NID)

ASMD (%)	TAW (mm)	NID (mm)
50	35.82	17.91

The left diagram shows calculation of NID determining by ASMD (Commonly set to 50%) and TAW

Case Study:

- The soil texture was identified as Sandy Loam with an AWC of 104 mm/m (Step 2).
- The orchard's durian trees are at Year 3, and the corresponding RZD is 34.44 cm (Step 3).
- From Step 4, the TAW was calculated as 35.82 mm.

Formula:

If ASMD is set at 50%, then:

$$\text{NID} = \text{TAW} \times \text{ASMD}$$

$$\text{NID} = 35.82 \text{ mm} \times 50\% = 17.91 \text{ mm}$$

Step 6:

Calculate Tree Canopy Area

The below diagram shows how to calculate the tree canopy area based on the canopy diameter and radius, which determined by the tree age.

Tree Age	(R) Canopy Diameter (m)	(R/2) Canopy Radius (m)	π	$\pi \times (R/2)^2$ Canopy Area (m ²)
3	3	1.5	3.14	7.065

Formula:

- Canopy Radius (R/2) = Canopy Diameter (R) divided by 2
- Canopy Area = $\pi \times (R/2)^2$

Case Study:

- For a 3 year old tree with a canopy diameter of 3 meter
- In this case, canopy area is 7.065 m²
- This canopy area will be used in Step 7 to calculate the water required per tree per day.

Remark:

To make it easier, number of tree age equals to canopy diameter.

Step 7:

Calculate Water Required per irrigation per tree per day

Tree Age	NID (mm)	$\pi*(R/2)^2$ Canopy Area (m ²)	(One Sprinkler) Water Required per irrigation per tree per day (L)	(Two Sprinkler) Water Required per irrigation per tree per day (L)
3	17.91	7.065	127	253

The left diagram shows calculation of water required per irrigation, determined by NID (Step 5) and Tree Canopy Area (Step 6). These values are adjusted based on one or two sprinklers

Formula:

Water Required per Tree = NID x Canopy Area

Case Study:

For a 3-year-old tree:

NID = 17.91 mm (from Step 5)

Canopy Area = 7.065 m² (from Step 6)

Water Required:

One Sprinkler =

$$17.91 \times 7.065 = 127 \text{ L/day}$$

Two Sprinklers =

$$17.91 \times 7.065 \times 2 = 253 \text{ L/day}$$

Disclaimer:

The calculations presented in this section are based on standardized formulas and serve as a general reference for estimating water requirements per tree per day. However, several factors such as evapotranspiration rates, climatic variations, and soil texture inconsistencies were not accounted for in this calculation. Additionally, the Available Water Capacity (AWC) values may vary with soil conditions, and the canopy area used here is an approximate estimate. These variables can significantly impact the actual water requirements, and adjustments may be necessary to suit specific orchard conditions. It is recommended to conduct regular field assessments to refine these estimates.

C. Overview of Water Requirement Calculations for Durian Trees (Year 1 – 10)

Tree Age	Soil Texture	AWC (mm/m)	RZD (cm)	TAW (mm)	ASM D (%)	NID (mm)	Canopy Diameter (m)	Canopy Area (m ²)	Water Required per tree per day (L)
1	Sandy Loam	104	30	31.2	50	15.6	1	0.79	12.25
2	Sandy Loam	104	32.22	33.51	50	16.75	2	3.14	52.61
3	Sandy Loam	104	34.44	35.82	50	17.91	3	7.07	126.53
4	Sandy Loam	104	36.66	38.13	50	19.06	4	12.56	239.43
5	Sandy Loam	104	38.88	40.44	50	20.22	5	19.63	396.77
6	Sandy Loam	104	41.1	42.74	50	21.37	6	28.26	603.97
7	Sandy Loam	104	43.32	43.05	50	22.53	7	38.47	866.48
8	Sandy Loam	104	45.54	47.36	50	23.68	8	50.24	1189.72
9	Sandy Loam	104	47.76	49.67	50	24.84	9	63.59	1579.15
10	Sandy Loam	104	49.98	51.98	50	25.99	10	78.5	2040.18

2.4.2 Fertilizer (Growing Stage)

This section provides guidelines for fertilizing durian trees during the vegetative phase (immature stage). This phase focuses on promoting healthy growth and development by supplying the necessary nutrients through conventional granular fertilizers. Proper fertilization during this stage supports root development, canopy expansion, and overall tree health, preparing the tree for the flowering and fruiting stage (reproductive phase).

This section applies to all durian trees during the vegetative phase (immature stage), defined as the period before the tree begins flowering and fruiting.

The immature stage typically spans the first to fifth year of the tree's growth, encompassing the vegetative phase, where the tree focuses on structural development and canopy formation.

For trees that begin flowering and fruiting earlier or later than the typical five-year period, this section applies until the tree enters the reproductive phase. Thereafter, refer to the **section 2.5.2 Fertilizer (Mature Stage)** for ongoing fertilization practices.

The below diagram shows the fertilization program from year 1 to 5 in LKE orchard.

Tree Age	Types of Fertilizer (N:P:K:Mg)	Amount of Fertilizer/ Tree in Kg	Number of Application per year
Year 1	15:15:6:4	0.2	12 (Monthly)
	Organic Fertilizer	5	4 (Quarterly)
Year 2	15:15:6:4	0.5	12 (Monthly)
	Organic Fertilizer	10	4 (Quarterly)
Year 3	15:15:6:4	1	12 (Monthly)
	Organic Fertilizer	15	4 (Quarterly)
Year 4	15:15:15	2	12 (Monthly)
	Organic Fertilizer	20	4 (Quarterly)
Year 5	12:12:17:2	2	12 (Monthly)
	Organic Fertilizer	25	4 (Quarterly)

2.4.3 Pest Management (Growing Stage)

This section provides guidelines for managing pests commonly encountered during the growing stage of durian trees. Effective pest management is critical to maintaining tree health, promoting canopy development, and ensuring optimal growth.

During this stage, pest control strategies focus on identifying and addressing specific pests that target the vegetative phase of the tree. Pests such as sucking or chewing pest can severely impact leaf structure, leading to defoliation and reduced efficiency of photosynthesis. As a result, the tree's ability to synthesize energy is compromised, ultimately affecting its overall health and productivity. In severe cases, prolonged pest infestation can stunt growth and significantly reduce fruit yield.

The pest management approach includes the use of targeted chemical insecticides and integrated pest control practices.

By combining these strategies, pest pressure is minimized while reducing the risk of resistance. For pests that affect trees during the harvesting stage, refer to the [section 2.5.3 Pest Management \(Mature Stage\)](#) for specific guidelines. Pest control during the harvesting stage focuses on protecting fruit quality and managing pests typically encountered during the reproductive phase.

The following sections will outline key pests identified during the growing stage, recommended solution treatments, and complementary control methods utilized in the LKE durian orchard.

A. Psyllid (*Allocaridara Malayensis*)

Causes sucking wounds that leave scars on the leaf surface (visible as red-circled marks). Primarily targets young leaves. While relatively easy to control, the species exhibits a rapid reproductive cycle leading to defoliation and reduced efficiency of photosynthesis of the durian tree.



The left diagram shows psyllid feeding on durian leaves. Below is a list of active ingredients used in the commercial pesticides to control psyllid

- Imidacloprid
- Pymetrozine
- Dinotefuran
- Neam oil
- Thiamethoxam + lambda-cyhalothrin
- Spiropidion + Acetamiprid

B. Hard Scale Insect (*Asterolecanium unguatum*)

Yellow, green, red, or black dots on the underside of leaves are female scale insects, which can be manually removed by scraping with a fingernail. In severe infestations, yellow specks appear on the leaf surface, indicating chlorophyll loss. Prolonged infestation may lead to significant foliage loss, resulting in a "balding" appearance and reduced efficiency of photosynthesis of the durian tree.

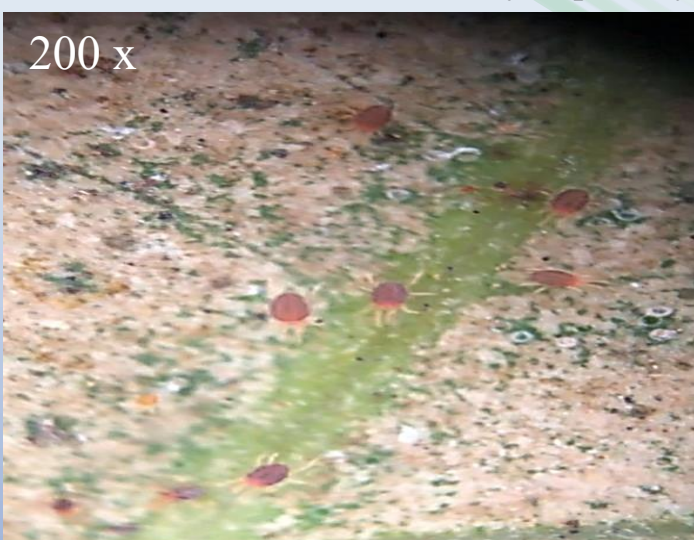


The left diagram shows durian hard scale insect feeding on durian leaves. Below is a list of active ingredients used in the commercial pesticides to control it.

- White Oil
- Methamidophos
- Thiamethoxam + lambda-cyhalothrin + pymetrozine
- Thiamethoxam + lambda-cyhalothrin + diafenthiuron
- Spiropidion + Acetamiprid + pymetrozine
- Spiropidion + Acetamiprid + diafenthiuron

C. Red Spider Mite (*Tetranychus urticae*)

Extract moisture from the leaves and young shoots of durian plants, leading to the desiccation of affected leaves. Infected leaves may eventually fall off, although they can sometimes remain green. The upper surface of the leaves often exhibits grayish specks that resemble dust leading to defoliation and reduced efficiency of photosynthesis of the durian tree.



The left diagram shows red spider mite feeding on durian leaves. Below is a list of active ingredients used in the commercial pesticides to control it.

- Abamectin
- Diafenthiuron

D. Stem Borer (Larvae of *Zeuzera coffeae*)

In durian orchards, stem borers are considered the number one threat among all pests. If you suddenly notice that some tree trunks or branches are drying out, breaking off, or even snapping entirely, there's a high chance that these pests are the cause.

Stem borers tunnel into the wood, disrupting the flow of water and nutrients. This leads to the affected parts drying out and dying due to dehydration. If not controlled in time, the infestation can spread and eventually kill the entire tree.

To manage the issue, it is crucial to immediately prune and remove affected branches. You can also inspect the cut sections for tunneling marks, which confirm the presence of stem borers and help assess the severity of the infestation.



The above diagram shows stem borer feeding in tree stems. Below is a list of active ingredients used in the commercial pesticides to control it.

- Emamectin-benzoate (IRAC 3A)
- Thiamethoxam + Chlorantraniliprole (IRAC 4A + 28)
- Lufenuron (IRAC 15)
- Thiamethozam + Lambda-cyhalothrin (IRAC 4A + 3A)
- Abamectin (IRAC 3A)
- Cypermethrin (IRAC 3A)

2.4.4 Disease Management (Growing Stage)

This section provides guidelines for managing diseases commonly encountered during the growing stage of durian trees. Effective disease management is critical to maintaining tree health, promoting canopy development, and ensuring optimal growth.

During this stage, disease control strategies focus on identifying and addressing specific disease that target the vegetative phase of the tree. Disease caused by fungal can severely impact leaf structure, leading to defoliation and reduced efficiency of photosynthesis. As a result, the tree's ability to synthesize energy is compromised, ultimately affecting its overall health and productivity. In severe cases, prolonged pest infestation can stunt growth and significantly reduce fruit yield.

The disease management approach includes the use of targeted fungicides and integrated disease control practices.

By combining these strategies, disease pressure is minimized while reducing the risk of resistance. For disease that affect trees during the harvesting stage, refer to the **section 2.5.4 Disease Management (Mature Stage)** for specific guidelines. Disease control during the harvesting stage focuses on protecting fruit quality and managing disease typically encountered during the reproductive phase.

The following sections will outline key diseases identified during the growing stage, recommended solution treatments, and complementary control methods utilized in the LKE durian orchard.

A. Stem Canker (*Phytophthora palmivora*)

Typically appears as dark, sunken, and blackened areas. When pressed, the infected area feels soft and exudes a liquid substance and weaken the tree's overall health. As the disease spreads, it disrupts nutrient transport, leading to branch dieback and eventually tree death. To manage this disease effectively, it is crucial to first remove the infected area by cutting away the affected bark and tissues. Only after this step can treatment be applied to prevent further spread and promote recovery.



The left diagram shows the infected spot caused by *P. palmivora*. Below is a list of active ingredients used in the commercial fungicides to control.

- Metalaxyl-M
- Fosetyl-Aluminium (*Preventive)
- Phosphorous Acid (*Preventive)
- Biocontrol agents like *Trichoderma harzianum*, *Chaetomium cupreum*, *C. gloeosporioides*, and *Bacillus subtilis* can be applied

B. Leaf Blight (*Rhizoctonia solani*)

Typically, water-soaked lesions begin on the leaves and progress upward along the stem. The coloration of stem lesions ranges from brown to black. In the later stages of the disease, affected leaves become so severely wilted that they can be easily detached from the plant leading to defoliation and reduced efficiency of photosynthesis of the durian tree.



The left diagram shows symptom of leaf blight on durian leaves. Below is a list of active ingredients used in the commercial fungicides to control it.

- Difenoconazole
- Azoxystrobin
- Pencycuron
- Carbendazim
- Mancozeb
- Propiconazole
- Copper oxychloride

C. Antracnose (*Colletotrichum gloeosporoides*)

Commonly affecting durian leaves, especially in humid and rainy conditions. Symptoms include irregular brown or black necrotic lesions with yellow halos, which may expand and cause leaf curling, drying, and defoliation. Severe infections can weaken the tree, reduce photosynthetic activity, and ultimately impact fruit yield.



The left diagram shows symptom of antracnose on durian leaves. Below is a list of active ingredients used in the commercial fungicides to control it.

- Difenoconazole
- Azoxystrobin
- Pencycuron
- Carbendazim
- Mancozeb
- Propiconazole

2.4.5 Weed Management (Growing Stage)

Effective weed management during the growing stage ensures optimal growth conditions for durian trees by minimizing competition for resources such as nutrients, water, and sunlight.

Mechanical Control: Regular manual or machine weeding is performed to maintain a clean orchard environment and prevent weeds from harboring pests and diseases. Frequent monitoring and weeding are essential due to the rapid regrowth of weeds, especially during the rainy season.

The following diagram shows a worker performing weed management using mechanical control methods in the LKE durian orchard.



2.4.6 Pruning (Growing Stage)

Pruning for young durian trees is focused on shaping the tree and removing unwanted growth to promote healthy structure and airflow. Key steps include:

- Removing new shoots that appear below the grafting point.
- Pruning lower branches up to approximately 70 cm to prevent unwanted growth close to the ground.
- Removing vertically growing branches that may interfere with the canopy structure.
- Eliminating water shoots and any unhealthy branches to ensure balanced growth.
- Pruning **should be** conducted at specific time (before fertilizer application) to optimize growth and tree health, prune water shoots to ensure nutrients are directed to essential parts of the tree.
- Pruning **should not be** conducted during or immediately after rainfall, as wet conditions increase susceptibility to diseases.

The following diagram shows Ideal Branch Structure and Arrangement for a Durian Tree.



2.5

Mature Stage (Harvest Stage)

2.5.1 Watering (Mature Stage)

This section provides guidelines for watering practices during the harvesting stage of durian cultivation.

The calculations for water requirements follow the framework and detailed steps outlined in [section 2.4.1](#).

For specific tree age (Year 1 to 4), refer to the overview table and the step-by-step calculation process in [section 2.4.1](#), which account for factors such as tree age, canopy size, and soil properties etc.

The following diagram provides an overview of the water requirements for durian trees from Year 5 to Year 10 during the harvesting stage, based on the calculations outlined in [section 2.4.1](#).

Tree Age	Soil Texture	AWC (mm/m)	RZD (cm)	TAW (mm)	ASMD (%)	NID (mm)	Canopy Diameter (m)	Canopy Area (m ²)	Water Required per tree per day (L)
5	Sandy Loam	104	38.88	40.44	50	20.22	5	19.63	396.77
6	Sandy Loam	104	41.1	42.74	50	21.37	6	28.26	603.97
7	Sandy Loam	104	43.32	43.05	50	22.53	7	38.47	866.48
8	Sandy Loam	104	45.54	47.36	50	23.68	8	50.24	1189.72
9	Sandy Loam	104	47.76	49.67	50	24.84	9	63.59	1579.15
10	Sandy Loam	104	49.98	51.98	50	25.99	10	78.5	2040.18

2.5.2 Fertilizer (Mature Stage)

This section provides guidelines for fertilizing durian trees during the mature stage, which begins as soon as the trees enter their first flowering and fruiting cycle.

The fertilization program during this stage supports the recovery of soil nutrients depleted during fruit development and harvest, promotes overall tree health and root development, and prepares the tree for subsequent flowering and fruiting cycles.

This section applies to all durian trees that have entered the reproductive phase, typically starting from Year 5. For trees that begin flowering and fruiting earlier or later than the typical five-year period, this section applies as soon as the first flowering and fruiting occur.

Refer to the [section 2.4.2 Fertilizer \(Growing Stage\)](#) for previous fertilization practices.

The below diagram shows the fertilization program from year 5 to 10 in LKE orchard.

Tree Age	Types of Fertilizer (N:P:K:Mg)	Amount of Fertilizer/ Tree in Kg	Number of Application per year
Year 5	12:12:17:2	2	12 (Monthly)
	Organic Fertilizer	25	4 (Quarterly)
Year 6	12:12:17:2	3	12 (Monthly)
	Organic Fertilizer	25	4 (Quarterly)
Year 7	12:12:17:2	3.5	12 (Monthly)
	Organic Fertilizer	25	4 (Quarterly)
Year 8	12:12:17:2	4	12 (Monthly)
	Organic Fertilizer	25	4 (Quarterly)
Year 9	12:12:17:2	4.5	12 (Monthly)
	Organic Fertilizer	25	4 (Quarterly)
Year 10	12:12:17:2	5	12 (Monthly)
	Organic Fertilizer	25	4 (Quarterly)

2.5.3 Pest Management (Mature Stage)

This section provides guidelines for managing pests commonly encountered during the mature stage of durian trees. Effective pest management is critical to maintaining tree health, promoting canopy development, and ensuring optimal growth.

During this stage, pest control strategies focus on identifying and addressing specific pests that target the fruit development of the tree. Pests such as sucking or chewing pest can severely impact leaf structure, leading to defoliation and reduced efficiency of photosynthesis. As a result, the tree's ability to synthesize energy is compromised, ultimately affecting its overall health and productivity. In severe cases, prolonged pest infestation can stunt growth and significantly reduce fruit yield.

The pest management approach includes the use of targeted chemical insecticides and integrated pest control practices.

By combining these strategies, pest pressure is minimized while reducing the risk of resistance. For pests that affect trees during the vegetative stage, refer to the [section 2.4.3 Pest Management \(Growing Stage\)](#) for specific guidelines. Pest control during the harvesting stage focuses on protecting fruit quality and managing pests typically encountered during the reproductive phase.

The following sections will outline key pests identified during the mature stage, recommended solution treatments, and complementary control methods utilized in the LKE durian orchard.

A. Mealy Bug (*Pseudococcus sp.*)

Mealybugs, aided by black ants, extract sap from young fruits, resulting in dark spots on the affected regions. Various species of mealybugs infest durian, primarily targeting the fruit and flowers, with rare occurrences on the leaves. Mealybugs excrete a sugary fluid that encourages the growth of sooty mold, which coats the fruit in a black layer. Durian fruits affected by both sooty mold and mealybugs appear unappealing, making them difficult to sell and resulting in lower market prices. Furthermore, the presence of fungi can inhibit photosynthesis, leading to poor plant growth.



The left diagram shows mealy bug feeding on durian fruits and leaves. Below is a list of active ingredients used in the commercial pesticides to control mealy bug

- Parathion
- Malathion
- Carbaryl
- Imidacloprid
- Thiamethoxam + lambda-cyhalothrin
- Spiropidion + Acetamiprid
- Diafenthiuron

B. Fruit Borer (*Conogethes punctiferalis*)

Fruit Borer are the immature forms of small moths characterized by yellow wings adorned with black spots. These pests target durian fruits of various sizes. The adult moth deposits its eggs on the exterior of the durian. Once the eggs hatched, the larvae begin consuming the pericarp of the fruit. As the larvae grow, they burrow into the fruit and feed on the flesh, leading to decay, or fruit rot.



The left diagram shows durian fruit borer feed on the fruit. Below is a list of active ingredients used in the commercial pesticides to control it.

- Emamectin-benzoate
- Abamectin
- Thiamethoxam + chlorantraniliprole
- Lufenuron
- Thiamethoxam + lambda-cyhalothrin
- Indoxacarb
- Carbosulfan



C. Seed Borer (*Mudaria luteileprosa*)

The mature moths lay their eggs on the durian fruit's surface, particularly near the stem. Once the larvae hatch, they penetrate the fruit to consume the seeds, leaving the durian flesh largely intact, aside from a minor entry point between the flesh and the pericarp.



The left diagram shows durian seed borer feed on the fruit. Below is a list of active ingredients used in the commercial pesticides to control it.

- Emamectin-benzoate
- Abamectin
- Thiamethoxam + chlorantraniliprole
- Lufenuron
- Thiamethoxam + lambda-cyhalothrin
- Indoxacarb
- Carbosulfan

A. Integrated Pest Management

In LKE, we adopt an Integrated Pest Management (IPM) approach, which combines multiple strategies to manage pests sustainably.

This approach reduces reliance on chemical pesticides and emphasizes the importance of cultural, mechanical, and biological practices to control pest populations effectively.

Key IPM Practice

- **Insect Light Traps:** Utilization of insect light traps with specific light spectrums to attract and capture harmful insects.
- **Pruning Practices:** Regular pruning of durian trees improves air circulation and reduces habitats for pests. This practice contributes to tree health and indirectly lowers pest infestations.
- **Weed Management:** Effective weed management minimizes alternate hosts for pests and disrupts their breeding cycles.
- **Reducing Chemical Reliance:** By integrating these practices, the reliance on chemical pesticides is significantly reduced. This not only lowers the risk of resistance development but also promotes a healthier orchard ecosystem.
- **Monitoring and Evaluation:** Regular monitoring of pest populations and the effectiveness of IPM strategies is conducted to ensure the approach remains adaptive and effective. Adjustments are made based on pest pressure and environmental factors.



**Daytime View of the
Insect Light Trap**



**Night-time View of
the Insect Light Trap**



**Moth Caught on the
Trap**

2.5.4 Disease Management (Mature Stage)

This section provides guidelines for managing diseases commonly encountered during the mature stage of durian trees. Effective disease management is critical to maintaining tree health, promoting canopy development, and ensuring optimal growth.

During this stage, disease control strategies focus on identifying and addressing specific disease that target the mature stage of the tree. The disease management approach includes the use of targeted chemical fungicides and integrated disease control practices.

Disease control during the harvesting stage focuses on protecting fruit quality and managing disease typically encountered during the reproductive phase.

The following sections will outline key diseases identified during the mature stage, recommended solution treatments, and complementary control methods utilized in the LKE durian orchard.

A. Fruit Rot (*Phytophthora palmivora*)

Infects durian fruit through its bottom, peduncle, and thorns, forming small lesions at these points. The conidia germinate into mycelium or hyphae within these lesions, which gradually enlarge until the hyphae become visible on the outer pericarp. Once symptoms of fruit rot are observed, immediate action is required to gather the infected fruit(s) and apply treatment to prevent spread. If left untreated, can affect large portion of the whole orchard yield quality in a short time.



The left diagram shows the infected spot caused by *P. palmivora*. Below is a list of active ingredients used in the commercial fungicides to control.

- Fosetyl-Aluminium (*Preventive)
- Phosphorous Acid (*Preventive)
- Dimethomorph (*Preventive)

2.5.5 Weed Management (Mature Stage)

Effective weed management during the mature stage ensures optimal growth conditions for durian trees by minimizing competition for resources such as nutrients, water, and sunlight.

Mechanical Control: Regular manual or machine weeding is performed to maintain a clean orchard environment and prevent weeds from harboring pests and diseases. Frequent monitoring and weeding are essential due to the rapid regrowth of weeds, especially during the rainy season.

The following diagram shows a worker performing weed management using mechanical control methods in the LKE durian orchard.



2.5.6 Pruning (Mature Stage)

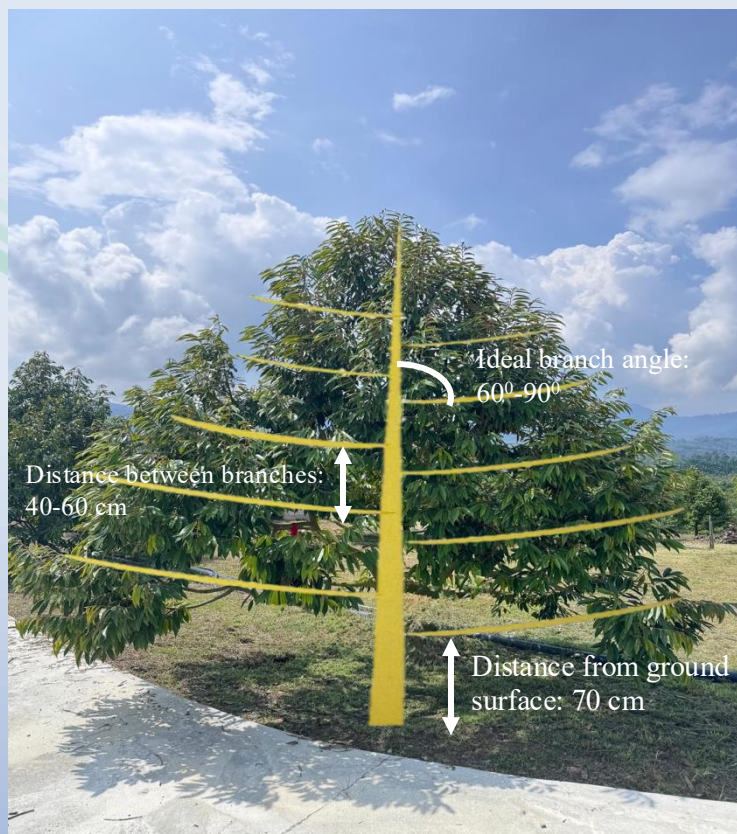
For mature or fruit-bearing durian trees, pruning is aimed at maintaining canopy structure, supporting fruit quality, and reducing disease risk. Important actions include:

- Removing water shoots and suckers that grow within the canopy, as these can hinder airflow and resource allocation.
- Pruning branches that have borne heavy fruit loads and may have weakened over time to prevent limb breakage.

Pruning **should be** conducted at specific times to optimize growth and tree health:

- Before fertilizer application: Prune water shoots to ensure nutrients are directed to essential parts of the tree.
- After fruit formation: Trim branches to support canopy management and prepare for future growth.
- Immediately after harvest: Remove dried or diseased branches to promote recovery and prepare for the next growth cycle.
- Pruning **should not be** conducted during or immediately after rainfall, as wet conditions increase susceptibility to diseases.

The following diagram shows Ideal Branch Structure and Arrangement for a Durian Tree.



2.5.7 Harvesting

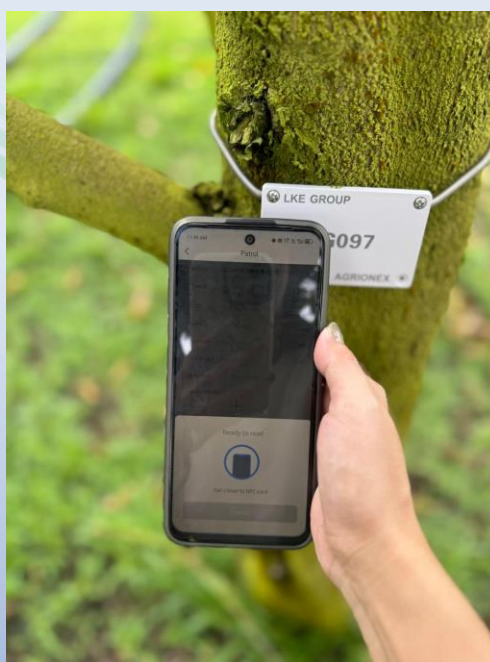
A. Pre-Harvest

- **Bundling:** Workers bundle durians that have reached the ideal size. The number of bundles and other details are recorded in the system using NFC tagging, which links directly to the respective tree ID.
- **Tracking:** The system tracks the number of bundled durians and their maturity stages, enabling effective monitoring of each tree's progress.
- **Managing Multiple Batches:** Unbundled durians (usually from the second or third batch) are monitored to reach the ideal size approximately two weeks after the first batch has harvested.

B. Post-Harvest

- **Collection and Weighing:** Workers collect fallen durians and weigh them. The weight and number of durians from each tree are recorded using NFC tagging, providing detailed yield analysis for each tree.
- **Grading and Labeling:** Durians are graded based on quality (e.g., A, B, C grades) and labeled according to their grade and harvesting phase.
- **Sales and Reporting:** Graded durians are sold to third-party buyers, with their weight and revenue recorded in the system. Revenue is calculated based on the current market price and the grade of the fruit.

Diagram below shows the actual scenario of NFC tagging in both pre-harvest and post-harvest stages, it forms one of the critical part of the **section 3.0 LKE Smart Farming Framework**.



3.0

LKE Smart Farming Framework

3.1

Environment Module

3.1.1 Weather

Our weather station continuously collects essential weather parameters, including:

- Rainfall
- Temperature
- Wind direction
- Wind speed
- Atmospheric pressure
- Humidity
- Illumination

These parameters are natural environmental factors that cannot be manually controlled or influenced. Instead, they are carefully monitored to provide a comprehensive understanding of the local climate conditions. The weather data collected every day is systematically stored and analyzed. This serves as the foundation for integrating advanced technologies such as AI and Machine Learning to generate predictive insights and optimize farming strategies.

The collected weather data forms the evidence-based standard for LKE Standard, representing the unique climatic conditions of our geographical location. These data-driven standards ensure that our practices are grounded in scientific observations rather than assumptions or estimates.

The following diagram shows example of daily weather data collected by our weather station.

Time	Ultrason ic Wind Speed (m/s)	Wind Direction (deg.)	Temperature (deg.)	Humidity (%)	Air Pressure (kPa)	Rainfall (mm)	Illumination (lux)
2024-12-30 08:44:07	7.2	120	28.5	78	1012.3	0	450
2024-12-30 08:32:34	6.8	130	28.2	80	1012.5	0	450
2024-12-30 08:21:05	6.6	135	27.9	81	1012.7	0	450

3.1.2 Soil

This section provides guidelines for soil sampling, soil sensor monitoring and implementing corrective actions based on data analysis and soil sampling results.

Both laboratory soil sampling and soil sensors play integral roles in maintaining optimal soil conditions. While soil sensors provide continuous, real-time monitoring to track long-term trends and detect immediate changes, laboratory soil tests offer detailed corrective recommendations based on precise data. By combining these two approaches—using lab results for targeted corrective actions and sensors for ongoing monitoring—we achieve a comprehensive and effective soil management strategy.

This integrated monitoring system ensures that corrective actions are data-driven and that soil health is consistently maintained to support tree growth and productivity.

A. Soil Sampling

Key Parameters for Analysis (Laboratory Analysis)

- Soil Structure: Percentage of sand, loam and clay.
- NPK (Nitrogen, Phosphorus, Potassium): Essential nutrients for plant growth.
- Magnesium (Mg): Vital for efficiency of photosynthesis and overall plant health.
- Electrical Conductivity (EC): Indicates soil salinity and helps assess the soil ability to retain nutrients.
- pH Level: A critical factor influencing nutrient availability; ideal pH levels for durian soil are necessary.
- Organic Carbon Content: Reflects soil organic matter, which contributes to soil fertility and water retention.

Testing Laboratories

- Soil samples should be sent to reputable institutions or government-certified laboratories for testing.
- These labs should have credibility and recognized standards to ensure accurate and reliable results.
- Soil sampling should follow consistent methodologies to guarantee comparability across tests.
- Annual testing is recommended to monitor and adjust soil management practices as needed.

Test Result

- After obtaining the test results from the laboratory, proceed recommended soil corrective actions if any parameter does not meet the standard range as recommended by the laboratory.
- These corrective actions will help to adjust soil conditions to meet optimal standards.

Soil Corrective Actions – Soil Sampling

- A detailed report will be generated by the laboratory.
- Act according to the laboratory’s recommendations, corrective actions are implemented to improve soil conditions. This may involve adjusting nutrient levels (N,P,K,Mg,S), soil pH, or organic matter content as specified in the lab report.
- After implementing the corrective actions, follow up soil sampling should be scheduled to evaluate the effectiveness of these adjustments.
- Continuous monitoring will help ensure that soil conditions remain optimal for plant health and productivity.

The following diagram shows example of soil sampling (pH) result report.

Sample Name	Soil pH	Interpretation & Recommendation	Recommended Lime Dosage by DOA (ton/ha)
Block 1	4.8	5 ton/ha of lime application is necessary	❖ pH > 5.5: No lime application required ❖ pH 5.0 – 5.5 : 2.5 ton of lime application required ❖ pH 4.5 – 5.0 : 5.0 ton of lime application required ❖ pH < 4.5 : 7.5 ton of lime application required DOA: Department of Agriculture
Block 2	5.1	2.5 ton/ha of lime application is necessary	
Block 3	4.3	7.5 ton/ha of lime application is necessary	

The following diagram shows example of soil sampling (EC) result report.

Sample Name	Soil EC (dS/m)	Interpretation & Recommendation	Note
Block 1	0.34	EC < 1, suitable for crop production	❖ 1 millimhos/cm = 1000 uS/cm ❖ 1 dS/m = 1000 uS/cm
Block 2	1.3	EC > 1, risk of salt stress and dehydration; do not over supply fertilizer	

Disclaimer:

The parameter and examples provided in this section, such as pH and EC, are for reference purposes only. Soil sampling methods, reporting format and recommended corrective actions may vary between laboratories, depending on their specific protocols and equipment. It is recommended to follow the guidelines and recommendations provided by the respective laboratory conducting the analysis to ensure accurate and reliable results.

B. Soil Sensor

The soil sensors provide continuous, real-time monitoring of soil conditions, enabling immediate insight into the current health and status of the soil. Unlike laboratory testing, which only provides data for a specific point in time, soil sensors allow ongoing observation and timely response to changing soil conditions.

Key Parameters for Monitoring (Soil Sensor)

- NPK (Nitrogen, Phosphorus, Potassium): Critical nutrients for tree growth and yield.
- EC (Electrical Conductivity): Helps assess soil salinity and nutrient retention.
- pH Level: Affects nutrient availability and is crucial for durian tree health.
- Soil Moisture Content: Tracks soil moisture levels, supporting proper irrigation practices.
- Soil Moisture Tension: Tracks root ability to uptake water from the soil.

Parameter Benchmarks

- Each parameter has an established recommended benchmark range (Refer to [section 2.1.1](#)).
- If values fall below or exceed these benchmarks, it may indicate potential issues that could negatively impact plant health.

*Diagram below shows the Summary of Soil Parameter Deficiency (Excessive) Potential Effect on Durian Trees.

	pH	EC	N	P	K	Moisture
Deficiency effect on Durian Tree	Risk of Nutrients Lockout	Risk of Lacking Soluble Nutrients	Risk of low photosynthesis rate lead to stunned growth	Risk of poor root development	Risk of having poor fruit quality	Risk of plant roots unable to absorb sufficient amount of water and lead to wilting
Excessive effect on Durian Tree	Risk of Nutrients Lockout	Risk of salt stress and dehydration	Risk of prone to disease and nitrogen toxicity	Risk of K imbalance	Risk of boron, calcium and magnesium imbalance	Risk of lead to poor aeration in the roots and water logging

Disclaimer:

The primary purpose of the soil sensor is to monitor trends in soil parameters over time, providing insights into changes in soil health. For precise recommendations on nutrient supplementation or corrective actions, users should rely on laboratory soil sampling reports and their specific recommendations. Additionally, it is important to note that many commercially available NPK sensors calculate nutrient levels using measured data combined with formulas, rather than directly measuring these values. Our sensor focuses on trend monitoring, emphasizing observation rather than direct supplementation guidance.

3.2 Cultivation Module

3.2.1 NFC-Based Task Classification

Individual Tree Task (Tasks are specific to individual tree and require NFC tapping on each tree after task completion.)

- Pruning
- Fertilizer Application
- Abnormality
- Pre-Harvest
- Post-Harvest
- Growth Monitoring
- Soil Testing
- Replanting

Block Based Task (A single NFC tap to any individual tree in the designated block will record the task completion for the entire area.)

- Pest & Control Spraying
- Irrigation
- Weed Management
- General Maintenance

3.2.2 Steps of Data Collection

1. Identify Task Type *To determine if the task is 2a(Single Tree) or 2b(Area-Based)

2a. Individual Tree Task

- Pruning
- Fertilizer Application
- Abnormality
- Pre-Harvest
- Post-Harvest
- Growth Monitoring
- Soil Testing
- Replanting

2b. Block Based Task

- Pest & Control Spraying
- Irrigation
- Weed Management
- General Maintenance

3a. Tap the NFC tag for the specific tree after completing the task. (*From 2a)

3b. Tap any NFC tag within the designated block to mark the task as complete for the entire designated area. (*From 2b)

4. All the data collected and sync to the system.

3.2.3 General Structure of Report Format

1. Single Tree Task Report Format

These tasks are specific to individual trees, so the reports should record detailed information for each task performed.

a. Pruning

- Tree ID: [NFC ID]
- Task Type: Pruning
- Date: [YYYY-MM-DD]
- Pruned Sections: [Optional to Upload pictures]
- Verified By: [Supervisor Name or ID]

b. Fertilizer Application

- Tree ID: [NFC ID]
- Task Type: Fertilizer Application
- Date: [YYYY-MM-DD]
- Fertilizer Type: [NPK ratio or organic fertilizer]
- Amount Applied: [Quantity, e.g., kg or g]
- Verified By: [Supervisor Name or ID]

c. Abnormality Report

- Tree ID: [NFC ID]
- Task Type: Abnormality Report
- Date: [YYYY-MM-DD]
- Abnormality: [Choose “YES” or “NO”]
- Observed Issues: [E.g., yellow leaves, pest infestations, upload photos]
- Action Taken: [E.g., pruning, pesticide application] *Until the abnormality becomes “NO”
- Verified By: [Supervisor Name or ID]

d. Pre-Harvest

- Tree ID: [NFC ID]
- Task Type: Pre-Harvest
- Date: [YYYY-MM-DD]
- Estimated Fruits Tied: [Number of fruits tied]
- Estimated Yield: [Weight in kg]
- Verified By: [Supervisor Name or ID]

e. Post-Harvest

- Tree ID: [NFC ID]
- Task Type: Post-Harvest
- Date: [YYYY-MM-DD]
- Harvested Fruits: [Number of fruits collected]
- Total Weight: [Weight in kg]
- Fruit Quality: [Grade breakdown, e.g., A, B, C]
- Verified By: [Supervisor Name or ID]

f. Growth Monitoring (*Related to Growth Module)

- Tree ID: [NFC ID]
- Task Type: Growth Monitoring
- Date: [YYYY-MM-DD]
- Tree Height: [Height in meters]
- Tree Circumference: [in cm]
- Tree Canopy Diameter: [in meters, if applicable]
- Condition: [Healthy/Needs Attention]
- Verified By: [Supervisor Name or ID]

g. Soil Testing

- Tree ID: [NFC ID or Block ID for area sampling]
- Task Type: Soil Testing
- Date: [YYYY-MM-DD]
- Sampling Location: [Tree or Block ID]
- Parameters Recorded: [e.g., N, P, K, pH, EC]
- Verified By: [Supervisor Name or ID]

h. Replanting

- Tree ID (Replanted): [NFC ID of the new tree]
- Task Type: Replanting
- Date: [YYYY-MM-DD]
- Tree Type/Variety: [Variety, e.g., Musang King, Black Thorn or others]
- Reason for Replanting: [Disease/Death/Empty Spot]
- Verified By: [Supervisor Name or ID]

2. Area-Based Task Report Format

These tasks apply to a specific area (block) and involve collective operations, so reports summarize the data for the entire area rather than individual trees.

a. Pest Control Spraying

- Block ID: [Block NFC ID]
- Task Type: Pest Control Spraying
- Date: [YYYY-MM-DD]
- Chemical Used: [Name of pesticide]
- Amount Used: [Quantity, e.g., liters]
- Verified By: [Supervisor Name or ID]

b. Watering

- Block ID: [Block NFC ID]
- Task Type: Watering
- Date: [YYYY-MM-DD]
- Water Amount: [Total water used, e.g., liters or cubic meters]
- Duration: [Duration of irrigation, e.g., hours]
- Verified By: [Supervisor Name or ID]

c. Weed Management

- Block ID: [Block NFC ID]
- Task Type: Weed Management
- Date: [YYYY-MM-DD]
- Method Used: [Mechanical]
- Coverage Area: [Size of area covered, e.g., acres or hectares]
- Verified By: [Supervisor Name or ID]

d. Others

- Block ID: [Block NFC ID]
- Task Type: General Maintenance
- Date: [YYYY-MM-DD]
- Details: [Specific maintenance tasks, e.g., upkeep boundary, cleaning drains, repairing roads]
- Verified By: [Supervisor Name or ID]

Disclaimer:

The cultivation module outlined in this section is designed with a strong focus on accurate and efficient data collection. By leveraging NFC technology, we ensure that every task performed on individual trees is systematically recorded, enabling precise tracking of tree-specific activities. This data-driven approach is critical for identifying inconsistencies, such as missed tree care, which would otherwise go unnoticed without reliable data.

To maintain the integrity of our data, the collection process has been simplified to eliminate unnecessary complexities. Overcomplicated steps can lead to inaccurate records, resulting in "Garbage In, Garbage Out" scenarios that compromise decision-making and orchard management. Our streamlined methods prioritize data accuracy, supporting a sustainable and efficient workflow for optimal orchard productivity.

3.3 Growth Module

The Growth Module focuses on monitoring and documenting the growth and health of durian trees. By leveraging NFC-based technology, this module ensures precise data collection, enabling effective analysis and informed decision-making for orchard management.

Key Components of Growth Module

3.3.1 Measurement of Tree Parameters Record

Records essential growth metrics, including tree height, trunk circumference, and canopy diameter. These measurements help track the development of each tree and identify any abnormalities requiring immediate attention. Data is collected through NFC tags linked to individual trees, ensuring accurate and consistent reporting.

3.3.2 Fruit Record

Tracks the number, weight, and quality (A, B, C grades) of fruits harvested from each tree. The module enables differentiation between estimated pre-harvest yields and actual post-harvest results, improving accuracy in yield forecasting and quality control. Collected data supports optimization of harvesting strategies and grading standards.

3.3.3 Growth Stage Record

Documents the development phases of trees, from seedling establishment to fruiting. This record ensures that all trees are progressing as expected and aligns management practices with the growth stage requirements. The module also helps in identifying growth trends and improving yield consistency..

3.3.4 Pest & Disease Record

Consolidates information on pest infestations and diseases observed in the orchard. Data collected through NFC tagging helps track affected areas and trees, facilitating targeted interventions like spraying or pruning. The module supports proactive orchard health management by identifying patterns and trends in pest and disease occurrences.

3.4 Finance Module

The Finance Module focuses on managing and analyzing the financial aspects of durian farming. By recording detailed data on costs and revenues, this module enables better budgeting, profitability tracking, and data-driven decision-making for long-term orchard management.

Key Components of Finance Module

3.4.1 Costing

Records all expenses incurred in orchard management, categorized into specific areas such as:

- **Fertilizer and Pesticides:** Tracks the quantity and cost of fertilizers, pesticides, and other inputs.
- **Labor Costs:** Documents expenses for tasks like pruning, harvesting, irrigation, and pest control.
- **Infrastructure Maintenance:** Includes expenses for irrigation systems, road maintenance, and general upkeep.
- **Operational Costs:** Captures other expenses, such as equipment, machinery, and transportation.

The module helps monitor and optimize spending, ensuring cost efficiency across all operations.

3.4.2 Revenue

Tracks income generated from durian sales, segmented by:

- **Fruit Quality:** Revenue is calculated based on the number and grade of fruits (e.g., A, B, C grades).
- **Market Price Trends:** Adjusts revenue projections based on fluctuating market prices.
- **Yield Per Tree:** Links income data to specific trees using NFC-tagged records, providing precise yield and revenue analysis.

Provides a comprehensive overview of profitability, allowing for better financial planning and forecasting.

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